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# USING MYSQL DBMS

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

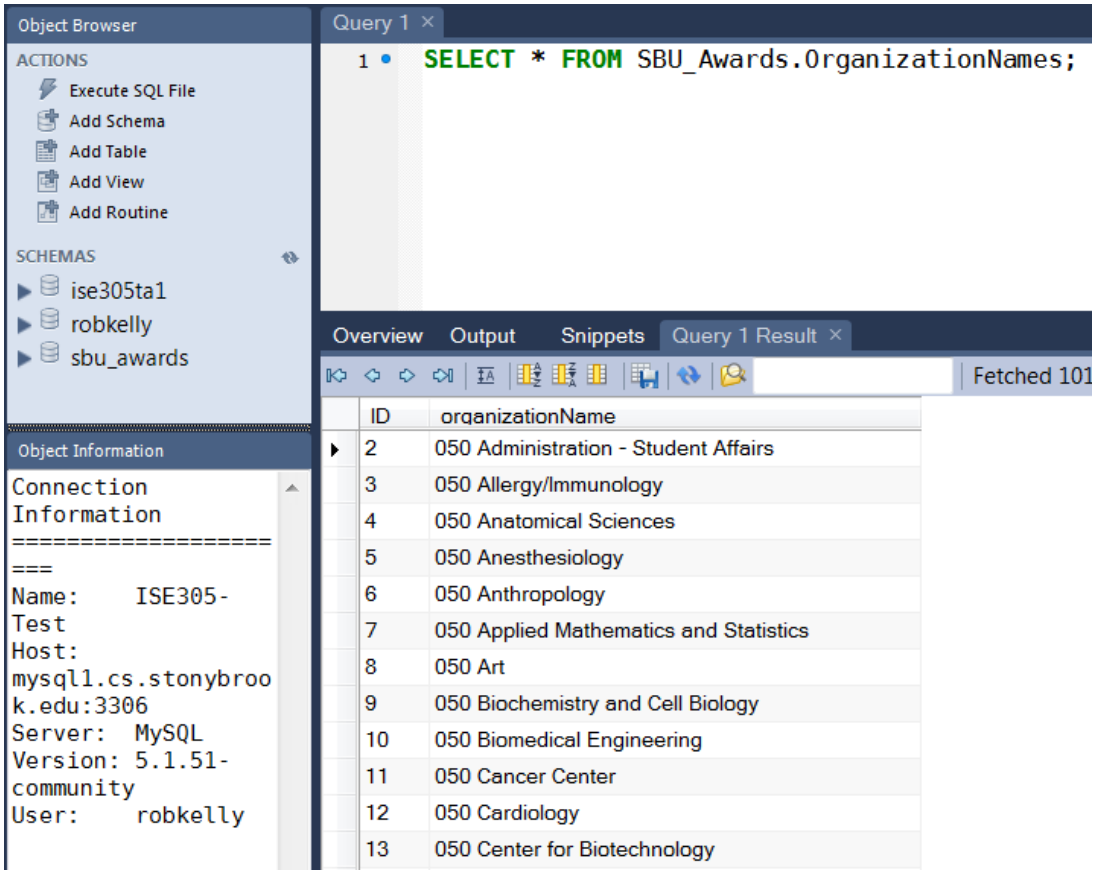
# Reading & Reference

- MySQL Usage Information

http://www.cs.stonybrook.edu/facilities/windowslab/services/mysql.html

# Workbench Access

- Workbench provides an interface to shared MySQL DBMS
- Start by connecting to the server



# MySQL Access

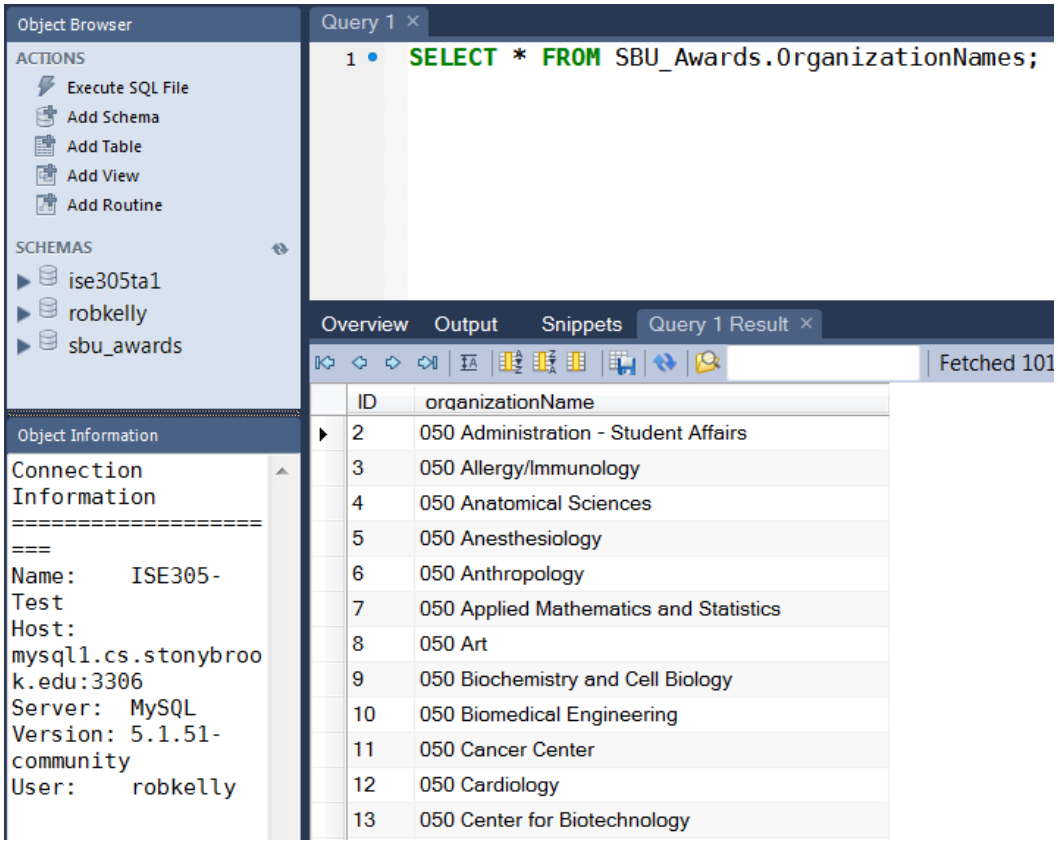
- An account is set for you in the MySQL server
- Your MySQL userid is the same as your NetID Your initial password of the database is your digit Stony Brook ID
- Your account has an empty database (database/schema name and the database userid are the same)

# MySQL Server Access

- Start MySQL Workbench
- Click on New Connection from SQL Development Module
- Specify the server as 'mysql1.cs.stonybrook.edu'
- Specify the port number as '3306'
- Specify the Username as your Net ID
- Specify your password as your SBID
- Specify your Default Schema as your Username.
- Click on OK

# Execute a Query

- Use the SQL pane in Workbench to execute a query



# Are We on Track?

- Run Workbench
- Connect to the server
- Access the SBU_Awards DB
- Execute a query to list the total budget of awards by sponsor

Overview Output Snippets Query 1 Result ×

Fetched 480 records. Durati

| sponsorName | totalBudget | numberAwards |
|---|---|---|
| National Science Foundation | 101197276.00 | 251 |
| US Department of Energy | 71826927.25 | 33 |
| Empire State Development Corp | 60914498.00 | 4 |
| National Institute of Allergy & Infectious Disease | 55867955.47 | 29 |
| National Institute of General Medical Sciences | 41472116.33 | 54 |
| Dormitory Authority of the State of New York | 36542000.00 | 2 |
| National Cancer Institute | 30104236.81 | 34 |
| National Inst of Neurological Disorders & Stroke | 19983384.67 | 22 |
| National Inst of Arthritis Musculoskeletal & Skin | 15570711.44 | 13 |
| NYS Office of Science Technology and Academic Res | 13795887.29 | 15 |
| National Oceanic and Atmospheric Administration | 13432094.00 | 10 |
| NYS Department of Health | 11471685.22 | 39 |

# Session Objectives

- Understand the background of the relational model
- Learn to
  - Model a single entity
  - Define a single database
  - Learn some SQL operations

# SQL Data Type Categories

- Character strings – includes fixed and variable length plus an option for international character sets
- Bit strings – fixed and variable length
- Integer (regular and short)
- Floating point (regular and double precision
- Numeric or decimal
- Date – includes date, time, date/time, plus timestamp and time zone

Look at the available data types in MS Access – and find better types for some SBU_Awards data

# Allowable Data Types

- SQL standard

| | | |
|---|---|---|
| Numeric | integer | A 31-bit signed binary value |
| | smallint | A 15-bit signed binary value |
| | float(*p*) | A scientific format number of *p* binary digits precision |
| | decimal(*p*,*q*) | A packed decimal number of *p* digits total length; *q* decimal places to the right of the decimal point may be specified |
| String | char(*n*) | A fixed length character string of *n* characters |
| | varchar(*n*) | A variable length character string up to *n* characters |
| | text | A variable-length character string of up to 65,535 characters |
| Date/time | date | Date in the form *yyyymmdd* |
| | time | Time in the form *hhmmss* |
| | timestamp | A combination of date and time to the nearest microsecond |
| | time with time zone | Same as time, with the addition of an offset from UTC of the specified time |
| | timestamp with time zone | Same as timestamp, with the addition of an offset from UTC of the specified time |

# Defining a Table in SQL

```
CREATE TABLE Shares(
   code        CHAR(3),
   firm        VARCHAR(20)NOT NULL,
   price       DECIMAL(6,2),
   qty         DECIMAL(8),
   div         DECIMAL(5,2),
   pe          DECIMAL(2),
       PRIMARY KEY(code));
```

We will not use SQL to define tables throughout most of the course

# Inserting Rows Using SQL

```
INSERT INTO Shares
  (code, firm, price, qty, div, pe)
 VALUES ('FC','Freedonia Copper',27.5,10529,1.84,16);
```

Or

```
INSERT INTO shr
 VALUES ('FC','Freedonia Copper',27.5,10529,1.84,16);
```

**Can also import from a text file**

# Did You Meet Session Objectives?

- Learn to
  - Model a single entity
  - Define a single database
  - Write queries for a single-table database